

AWS VPC - Study Notes

1. What is VPC?

VPC (Virtual Private Cloud) is a private network inside AWS. It controls how AWS resources communicate with each other and the internet.

Key Points: Region-based, contains subnets across AZs, defines IP range (CIDR), routing, and security.

2. Subnet

A smaller IP range inside a VPC. Each subnet belongs to exactly **one Availability Zone**.

Example: VPC 10.0.0.0/16 contains Subnet 1: 10.0.1.0/24, Subnet 2: 10.0.2.0/24

3. Internet Gateway (IGW)

Provides internet connectivity to a VPC. A subnet is **public** only if its route table has a route to IGW.

4. Route Table

Decides where network traffic should go. Format: **Destination** → **Target**

Example: 10.0.0.0/16 → local (stays in VPC) | 0.0.0.0/0 → IGW (goes to internet)

5. Public vs Private Subnet

Public Subnet: Has route to IGW. Used for web servers, load balancers.

Private Subnet: No direct IGW route. Used for databases, app servers, internal services.

6. NAT Gateway

Allows private subnet resources to make **outbound** internet requests without being directly reachable from internet.

Created in public subnet. Private resources use it to access external APIs, download updates, etc.

7. Simple VPC Architecture

```
Internet | Internet Gateway (IGW) | ████████████████████████████████████ | VPC:  
10.0.0.0/16 | | Public Subnet | | 10.0.1.0/24 | | - Web Server | | - NAT Gateway | |  
Private Subnet | | 10.0.2.0/24 | | - App Server | | - Database |  
████████████████████████████████████████
```

Traffic Flow

Inbound: Internet → IGW → Public Subnet

Outbound (Private): Private Subnet → NAT Gateway → IGW → Internet

Quick Reference

VPC: Virtual network in AWS

Subnet: IP range in one AZ

IGW: Connects VPC to internet

Route Table: Directs traffic

Public Subnet: Has IGW route

Private Subnet: No IGW route

NAT Gateway: Outbound internet for private subnets